

Data Pipeline Summary

Practice : small Flight Pipeline and Dashboard Project

✓ Introduction:

A **data pipeline** automates the process of transforming raw data into clean, structured, business-ready tables. Instead of manually running scripts or notebooks every day, pipelines ensure data is continuously ingested, transformed, and ready for analysis or machine learning.

Key benefits:

- Reduces manual errors
- Ensures fresh, reliable data
- Simplifies complex orchestration
- Supports analytics, dashboards, and ML workflows

✓ Core Concepts:

2.1 Pipelines as a Factory Line

Think of a pipeline as a factory:

- **Inputs:** Raw files, logs, API data, or streaming sources
- **Middle:** Cleaning, transformations, joins, aggregations
- **Outputs:** Analytics tables, dashboards, ML-ready datasets

This model ensures **predictable, repeatable, and maintainable data processing**.

2.2 Declarative Pipelines

Declarative pipelines focus on **what the final tables should look like**, not on how to orchestrate steps manually.

Benefits:

- Reduces custom orchestration code
- Databricks automatically determines execution order
- Handles incremental updates efficiently
- Ensures reliable, always-fresh data

2.3 The Medallion Architecture

Organizes data into layers for clarity and scalability:

Layer	Description
Bronze	Raw, ingested data
Silver	Cleaned and structured data
Gold	Aggregated business-level tables

Each layer **builds on the previous**, enabling reliable, incremental data flows.

✓ Pipeline Components:

3.1 Ingestion (Bronze Layer)

- Pulls raw data from APIs (like OpenSky), files, or streaming sources
- Handles schema inference and incremental updates
- Locally, we implemented this as `ingest_flights.py` reading flight data into Spark

3.2 Transformation & Cleaning (Silver Layer)

- Cleans and structures raw data
- Removes duplicates, handles missing values
- Example functions in this project:
 - `clean_flights(df)`
 - `flight_stats(df)`
 - `aircraft_by_country(df)`
 - `fastest_flights(df)`
 - `flights_on_ground(df)`

3.3 Aggregation & Business Metrics (Gold Layer)

- Produces business-ready tables for dashboards or ML
- Aggregates statistics such as:
 - Total flights
 - Distinct aircraft
 - Max velocity
 - Flights by country
 - Fastest flights
 - Flights on the ground
- Saved as **Parquet tables** for efficient querying

3.4 Pipeline Runner

- Executes the steps sequentially:
 - Ingest
 - Clean
 - Transform
 - Save results
- Supports **streaming simulation** with a loop and `time.sleep()` for continuous updates

❏ Dashboard Integration:

- Built with **Streamlit** and **Plotly**
- Displays:
 - KPIs (total events, distinct aircraft, max velocity)
 - Bar chart: aircraft by country
 - Top 10 fastest flights
 - Flights on the ground
 - Interactive world map of aircraft positions
- Connects directly to **pipeline outputs**, showing near real-time data

✓ Lessons Learned:

By completing this project, you should now understand:

1. How **pipelines automate end-to-end data processing**
2. How the **Medallion Architecture (Bronze → Silver → Gold)** organizes data
3. How to implement **ingest, transform, and aggregation steps** in PySpark
4. How to **store results** for analytics and dashboards
5. How to use **Streamlit + Plotly** to build interactive dashboards
6. The value of **declarative pipelines**:
 - a. Focus on logic, not orchestration
 - b. Reduce errors and maintenance
 - c. Enable continuous, incremental processing
7. Practical skills:
 - a. Spark DataFrame transformations
 - b. Aggregations and filtering
 - c. Writing to Parquet for incremental pipelines
 - d. Visualizing results interactively

Optional Advanced Extensions:

To make the project more **portfolio-ready**, you could add:

- ✓ **Real-time streaming** using Spark Structured Streaming
- ✓ **Auto-refreshing dashboard** for live aircraft movement
- ✓ **Air traffic heatmaps** and analytics
- ✓ **Historical trends** and data versioning
- ✓ Integration with **Kafka** or **cloud storage** for scalable ingestion

✓ Key Takeaways:

- A pipeline is an automated workflow turning raw data into analytics-ready tables
- Declarative pipelines simplify engineering and reduce manual overhead
- Medallion architecture organizes your data for reliability and scalability
- Dashboards depend on the Gold layer for accurate, fresh insights
- Automating ingestion, transformation, and visualization **creates a production-ready workflow**